

The whole lattice map and its prime fibres

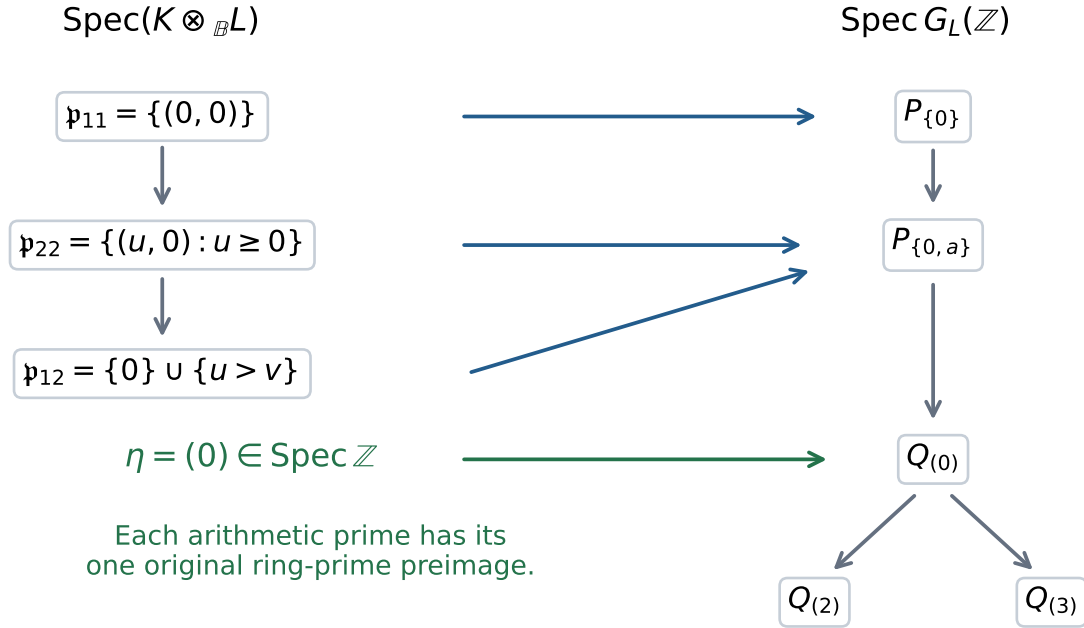
$L = \{0 < a < 1\}$, with join and meet; $K = ([0, \infty), \max, \cdot)$.

$$G_L(\mathbb{Z}) \longrightarrow \mathbb{Z} \times (K \otimes_{\mathbb{B}} L)$$

$$(r, \lambda) \mapsto (r, \mathbf{1}_{\{j \leq \lambda\}})$$

$\tau = z_0$	$\longrightarrow$	$(0; (0, 0))$
z_a	$\longrightarrow$	$(0; (1, 0))$
$e = z_1$	$\longrightarrow$	$(0; (1, 1))$
$\hat{n} \quad (n \neq 0)$	$\longrightarrow$	$(n; (1, 1))$

Prime contraction: the complete fibres



Grey arrows: strict prime inclusion. Blue/green arrows: contraction.
Only primes 2 and 3 are drawn; every rational prime remains.

The additional prime is retained

$$(2, 1) \in \mathfrak{p}_{12}, \quad (2, 1) \notin \mathfrak{p}_{11} \cup \mathfrak{p}_{22}.$$

$$\text{Fr}_0^{-1}(\mathfrak{p}_{12}) = \mathfrak{p}_{22}, \quad \text{Fr}_t^{-1}(\mathfrak{p}_{ij}) = \mathfrak{p}_{ij} \quad (t > 0).$$

Proofs: ASL18–22, ASL31; LSP3–27. Tropical coefficients: Connes–Consani, Geometry of the Arithmetic Site, arXiv:1502.05580v1, Definition defnpt.
Full lattice base: The Clankers, Split Support Geometry v11, def:lattice-split.